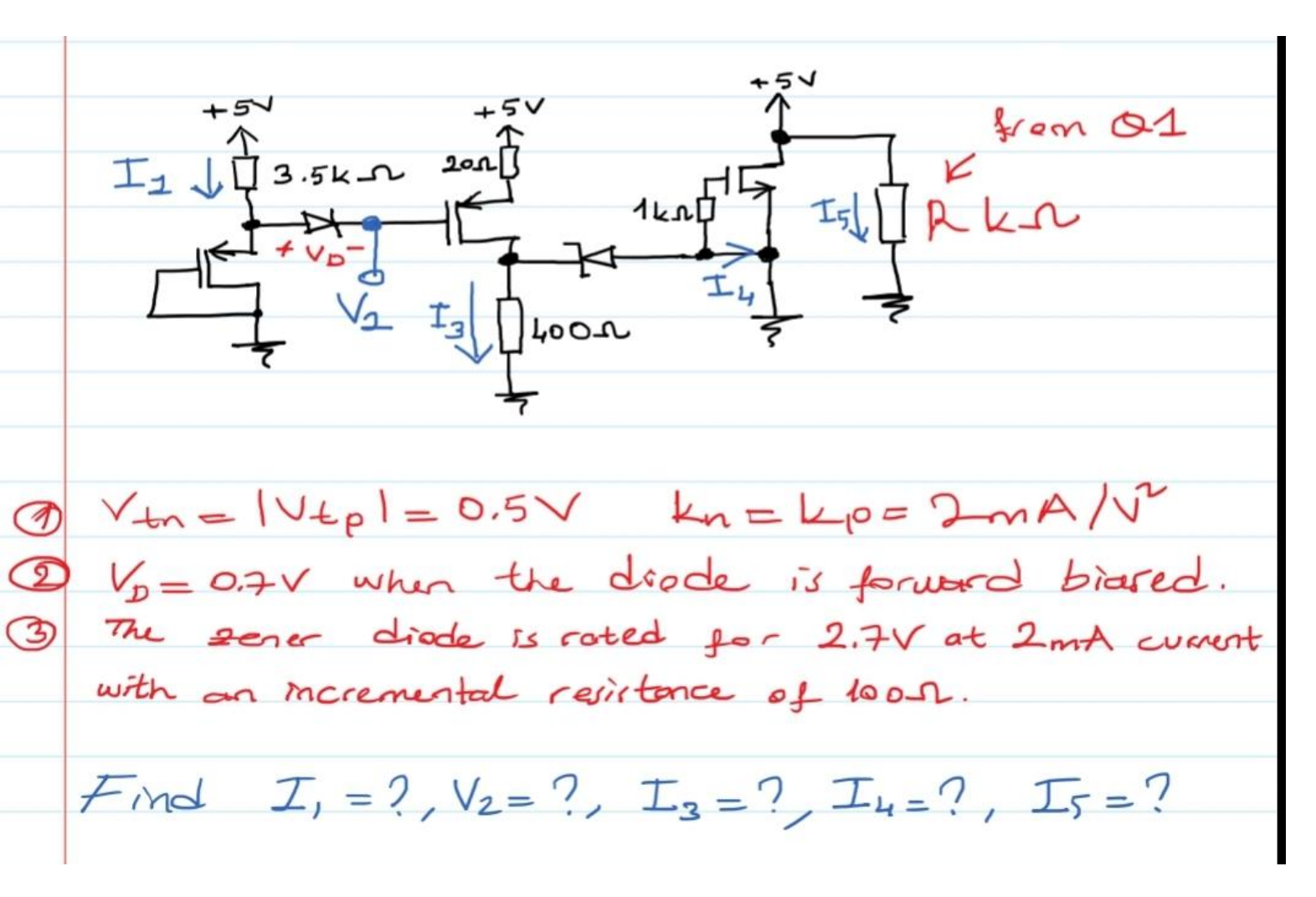
Question: +5 +5V +5V frem ol I 3.5kr 2010 1krd Is Irkon ----- Iu I Droon ② t…

R=10



① $V_{tn} = |V_{tp}| = 0.5V$ $k_n = k_p = 2mA/V^2$

② $V_D = 0.7V$ when the diode is forward biased.

③ The zener diode is rated for 2.7V at 2mA current with an incremental resistance of 100Ω.

Find $I_1 = ?, V_2 = ?, I_3 = ?, I_4 = ?, I_5 = ?$

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Expert Answer

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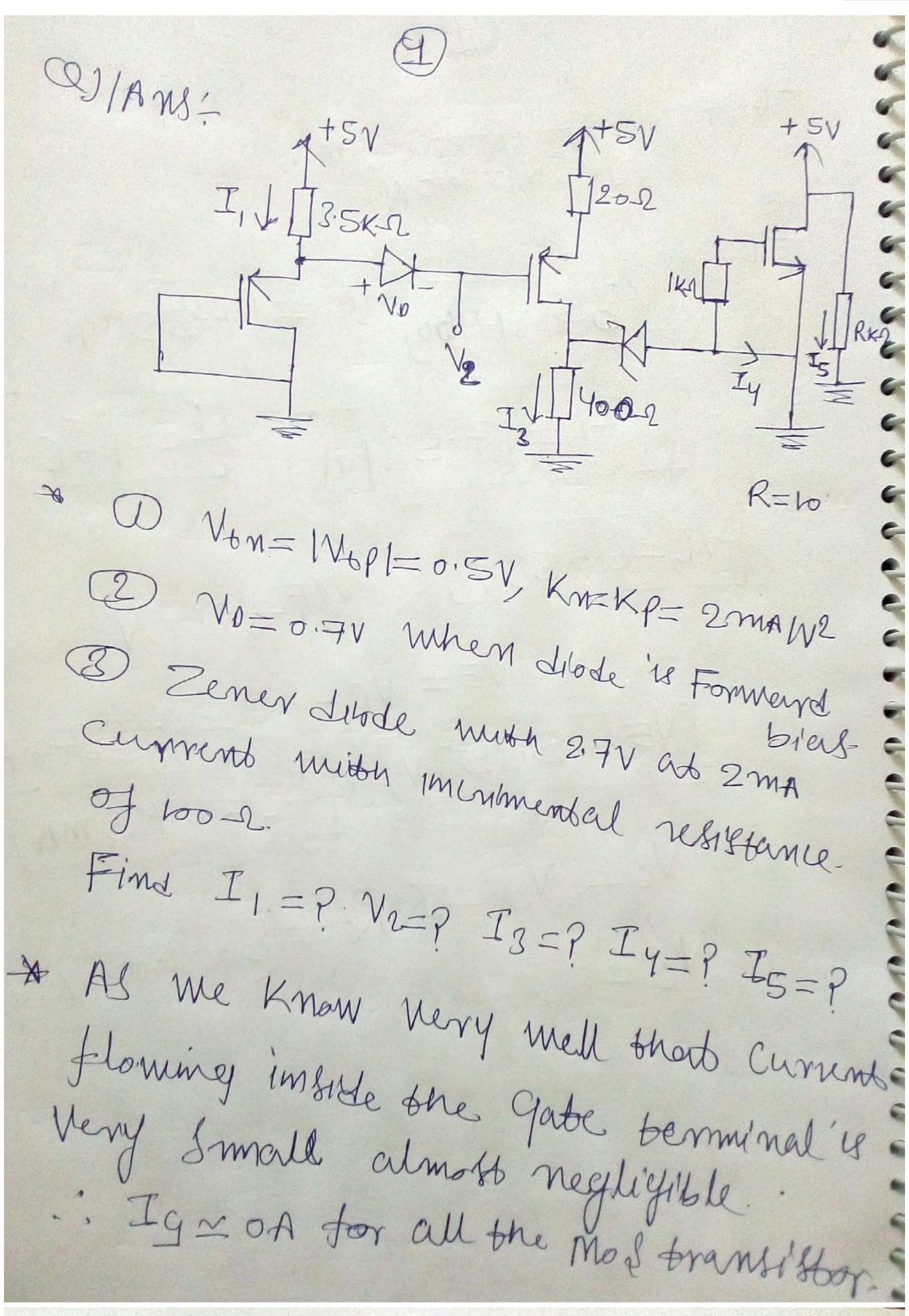

+5V
I_1
3.5KΩ
+5V
20Ω
+5V
V_D
V_2
1KΩ
RKΩ
I_5
I_4
I_3
400Ω

①

Q/Ans:-

R=10

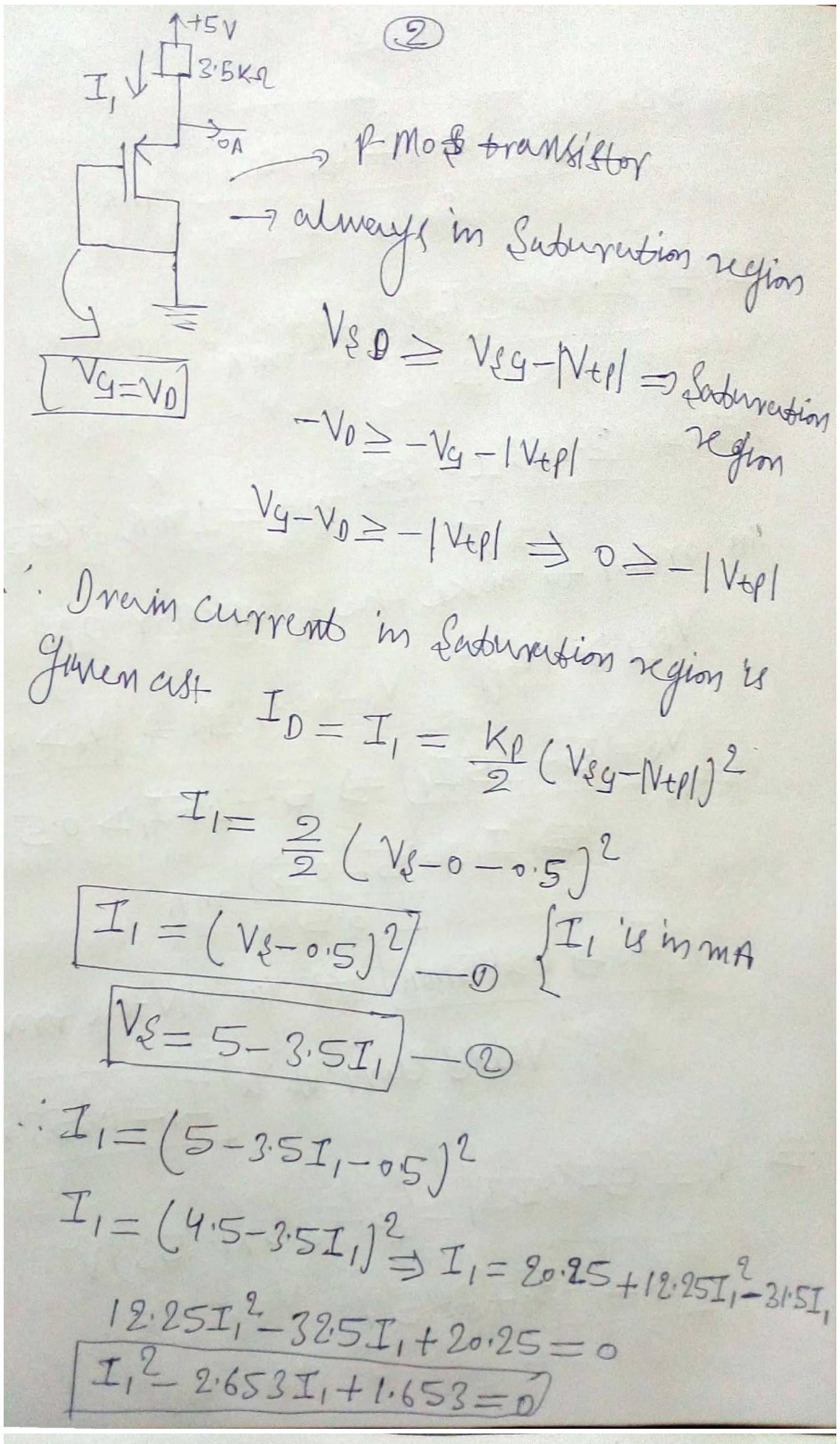
① $V_{tn} = |V_{tp}| = 0.5V$, $K_n = K_p = 2mA/V^2$

② $V_D = 0.7V$ when diode is Forward bias

③ Zener diode with 2.7V at 2mA current with incremental resistance of 100Ω.

Find $I_1 = ?$ $V_2 = ?$ $I_3 = ?$ $I_4 = ?$ $I_5 = ?$

* As we know very well that current flowing inside the gate terminal is very small almost negligible.
∴ $I_G \simeq 0A$ for all the MoS transistor.


+5V
3.5KΩ
I_1
0A

②

$\boxed{V_G = V_D}$

→ P-MoS transistor
→ always in Saturation region

$V_{SD} \geq V_{SG} - |V_{tp}| \Rightarrow$ Saturation region

$-V_D \geq -V_G - |V_{tp}|$

$V_G - V_D \geq -|V_{tp}| \Rightarrow 0 \geq -|V_{tp}|$

∴ Drain current in saturation region is given as:-

$I_D = I_1 = \frac{K_P}{2}(V_{SG} - |V_{tp}|)^2$

$I_1 = \frac{2}{2}(V_S - 0 - 0.5)^2$

$\boxed{I_1 = (V_S - 0.5)^2}$ — ① { I_1 is in mA

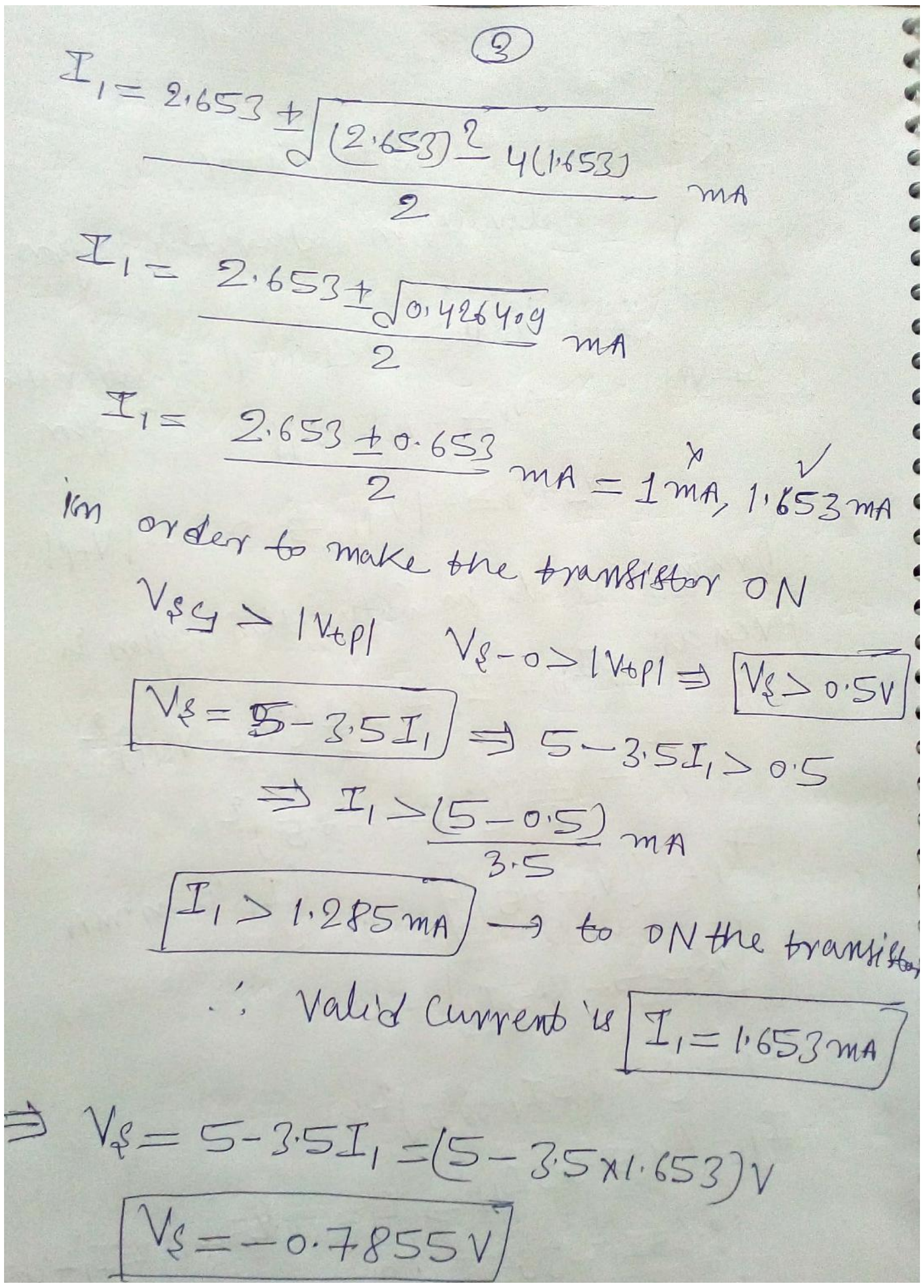
$\boxed{V_S = 5 - 3.5 I_1}$ — ②

∴ $I_1 = (5 - 3.5I_1 - 0.5)^2$

$I_1 = (4.5 - 3.5I_1)^2 \Rightarrow I_1 = 20.25 + 12.25I_1^2 - 31.5I_1$

$12.25I_1^2 - 32.5I_1 + 20.25 = 0$

$\boxed{I_1^2 - 2.653I_1 + 1.653 = 0}$



③

$I_1 = \frac{2.653 \pm \sqrt{(2.653)^2 - 4(1.653)}}{2}$ mA

$I_1 = \frac{2.653 \pm \sqrt{0.426409}}{2}$ mA

$I_1 = \frac{2.653 \pm 0.653}{2}$ mA $= \overset{\times}{1mA}, \overset{✓}{1.653mA}$

In order to make the transistor ON

$V_{SG} > |V_{tp}|$ $V_S - 0 > |V_{tp}| \Rightarrow \boxed{V_S > 0.5V}$

$\boxed{V_S = 5 - 3.5I_1} \Rightarrow 5 - 3.5I_1 > 0.5$

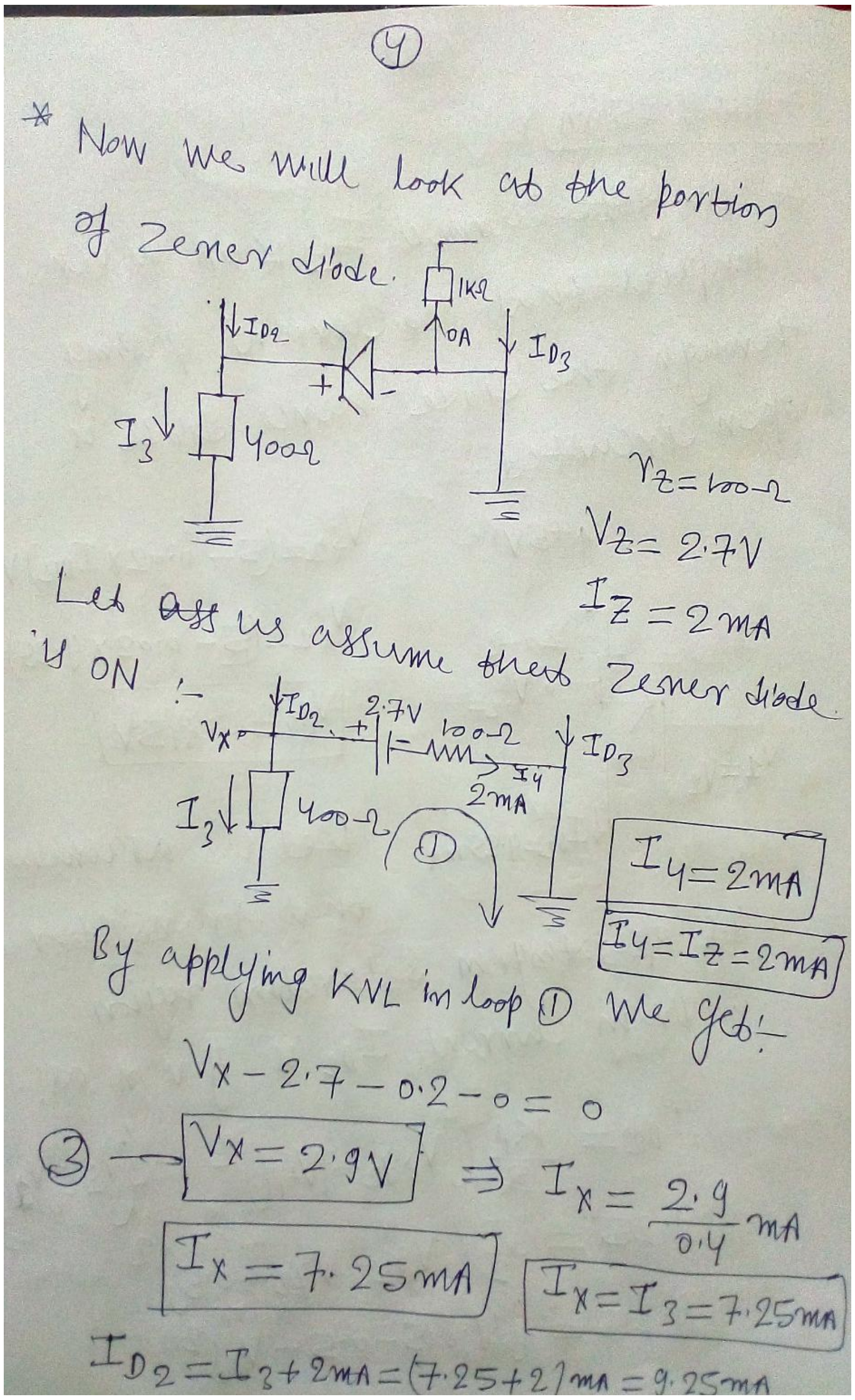
$\Rightarrow I_1 > \frac{(5 - 0.5)}{3.5}$ mA

$\boxed{I_1 > 1.285mA}$ → to ON the transistor

∴ Valid current is $\boxed{I_1 = 1.653mA}$

⇒ $V_S = 5 - 3.5I_1 = (5 - 3.5 \times 1.653)V$

$\boxed{V_S = -0.7855V}$


1KΩ
I_{D2}
0A
I_{D3}
I_3
400Ω
V_x
I_{D2}
2.7V
100Ω
I_{D3}
I_4
2mA
I_3
400Ω
①

④

* Now we will look at the portion of Zener diode.

$r_z = 100Ω$
$V_Z = 2.7V$
$I_Z = 2mA$

Let ~~ass~~ us assume that Zener diode is ON :-

$\boxed{I_4 = 2mA}$

$\boxed{I_4 = I_Z = 2mA}$

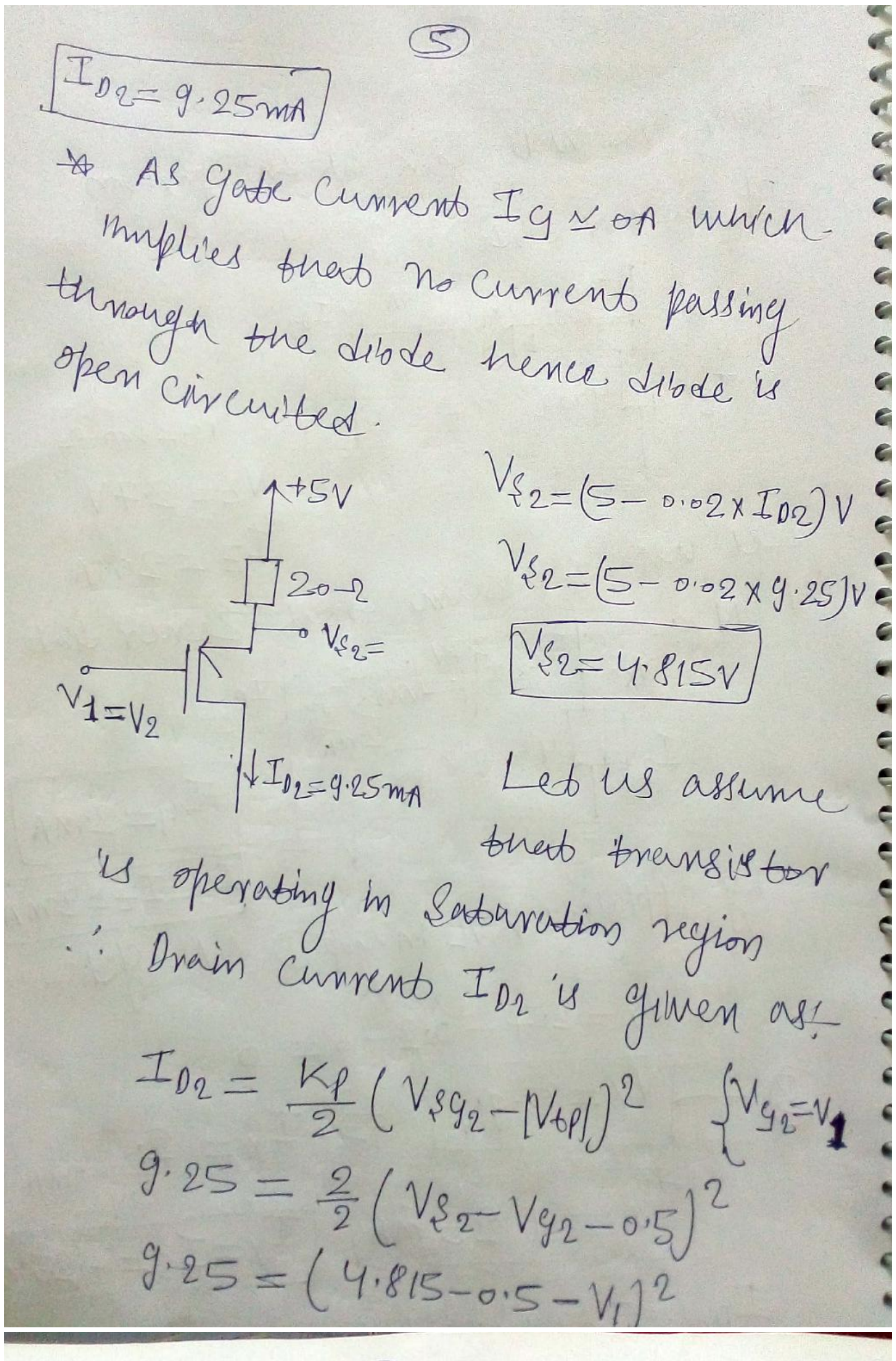
By applying KVL in loop ① we get:-

$V_x - 2.7 - 0.2 - 0 = 0$

③ — $\boxed{V_x = 2.9V} \Rightarrow I_x = \frac{2.9}{0.4}$ mA

$\boxed{I_x = 7.25mA}$ $\boxed{I_x = I_3 = 7.25mA}$

$I_{D2} = I_3 + 2mA = (7.25 + 2)mA = 9.25mA$


+5V
20Ω
$V_{S2}=$
$V_1 = V_2$
$I_{D2} = 9.25mA$

⑤

$\boxed{I_{D2} = 9.25mA}$

* As gate current $I_G \simeq 0A$ which implies that no current passing through the diode hence diode is open circuited.

$V_{S2} = (5 - 0.02 \times I_{D2})V$

$V_{S2} = (5 - 0.02 \times 9.25)V$

$\boxed{V_{S2} = 4.815V}$

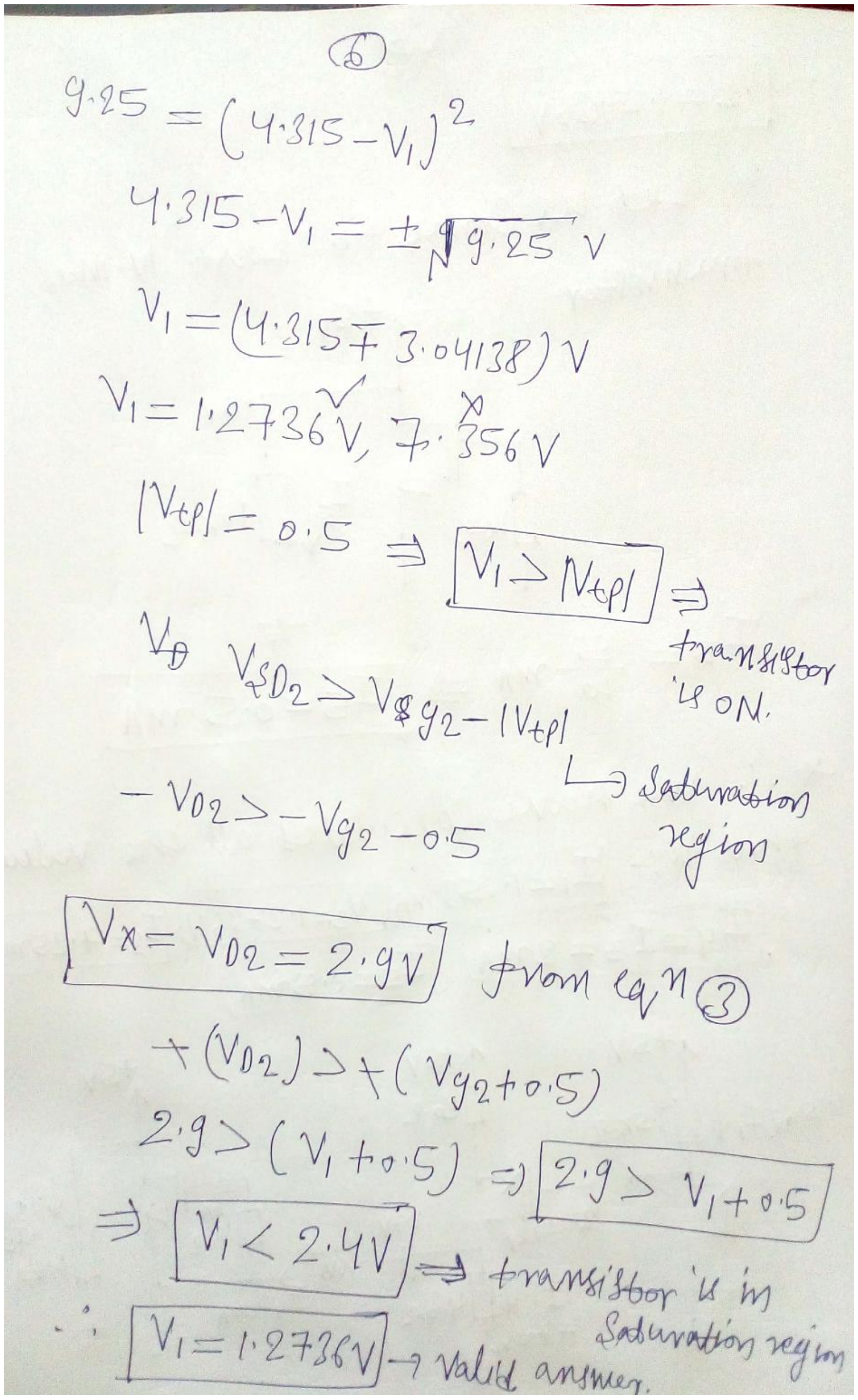
Let us assume that transistor is operating in Saturation region

∴ Drain current I_{D2} is given as:-

$I_{D2} = \frac{K_P}{2}(V_{SG2} - |V_{tp}|)^2$ { $V_{G2} = V_1$

$9.25 = \frac{2}{2}(V_{S2} - V_{G2} - 0.5)^2$

$9.25 = (4.815 - 0.5 - V_1)^2$



⑥

$9.25 = (4.315 - V_1)^2$

$4.315 - V_1 = \pm\sqrt{9.25}$ V

$V_1 = (4.315 \mp 3.04138)$ V

$V_1 = \overset{✓}{1.2736V}, \overset{\times}{7.356V}$

$|V_{tp}| = 0.5 \Rightarrow \boxed{V_1 > |V_{tp}|} \Rightarrow$ transistor is ON.

~~V_D~~ $V_{SD2} > V_{SG2} - |V_{tp}|$
↳ Saturation region

$-V_{D2} > -V_{G2} - 0.5$

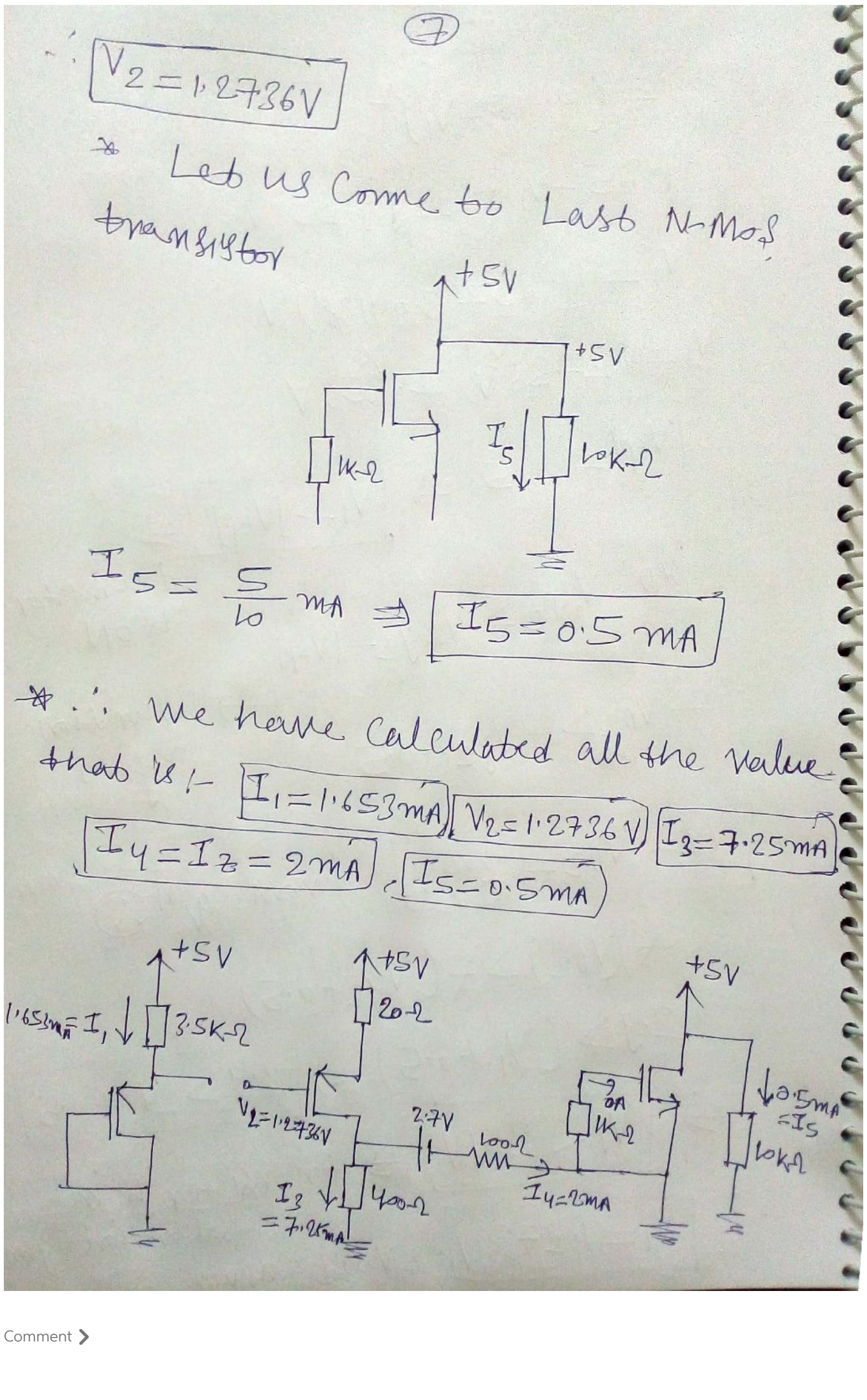
$\boxed{V_x = V_{D2} = 2.9V}$ from eqn ③

$+(V_{D2}) > +(V_{G2} + 0.5)$

$2.9 > (V_1 + 0.5) \Rightarrow \boxed{2.9 > V_1 + 0.5}$

$\Rightarrow \boxed{V_1 < 2.4V} \Rightarrow$ transistor is in Saturation region

∴ $\boxed{V_1 = 1.2736V}$ → Valid answer.


+5V
+5V
1KΩ
I_5
10KΩ
+5V
$1.653mA = I_1$
3.5KΩ
+5V
20Ω
$V_2 = 1.2736V$
2.7V
100Ω
$I_3 = 7.25mA$
400Ω
$I_4 = 2mA$
0A
1KΩ
+5V
$0.5mA = I_5$
10KΩ

⑦

∴ $\boxed{V_2 = 1.2736V}$

* Let us come to Last N-MoS transistor

$I_5 = \frac{5}{10}$ mA $\Rightarrow \boxed{I_5 = 0.5mA}$

* ∴ We have calculated all the value that is:- $\boxed{I_1 = 1.653mA}$ $\boxed{V_2 = 1.2736V}$ $\boxed{I_3 = 7.25mA}$ $\boxed{I_4 = I_Z = 2mA}$ $\boxed{I_5 = 0.5mA}$

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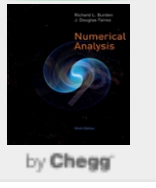
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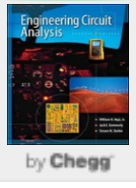
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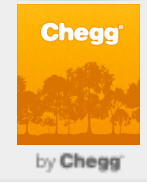
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